🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 · STANDARD 6.DS.5

Represent and Describe Data in a Box Plot

Lesson 2-4 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can make and read a box plot to summarize a data set.
Language Objective: I can explain my box plot using the words box plot, median, quartile, and interquartile range.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Box Plot | Median | Quartile | Interquartile Range | Data distribution | Variability

👆 *Tap any word to see its meaning and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. A graph that shows data with a box and whiskers based on the five-number summary is a ______________.

2. The middle value of the data, shown as a line inside the box, is the ______________.

3. A value that divides the data into four equal parts is a ______________.

4. The distance between the first and third quartiles, showing the middle 50% of data, is the ______________.

5. The way data values are spread out is the ______________.

6 How spread out the data values are is the ______________________.

Write About the Math

Which team is more consistently scoring high? Use the box plot values to support your answer?

¿Qué equipo anota alto de forma más consistente? Usa los valores del diagrama de caja para apoyar tu respuesta?

Say your answer to a partner first. Then write it, using at least two of these words:

- [ ] **Box Plot** *Diagrama de caja*
- [ ] **Median** *Mediana*
- [ ] **Quartile** *Cuartil*
- [ ] **Interquartile Range** *Rango intercuartílico*
- [ ] **Data distribution** *Distribución de datos*

Model: *A box plot is built from the minimum, Q1, median, Q3, and maximum.* — Students name the five-number summary (min, Q1, median, Q3, max) and find the median (14) for the 11 ordered scores.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Team A's IQR is ____ and Team B's IQR is ____. Team B's middle 50% scores between ____ and ____. The fan is wrong because ____. The box plot shows that Team B is more consistent because its Q1 and median are higher and its IQR (10) is smaller, while ____.**
- **My answer is ____ because ____.**

  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

  Mi afirmación es ____.
- **My evidence is ____, so ____.**

  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Box Plot
2. **Notes 2:** Median
3. **Notes 3:** Quartile
4. **Notes 4:** Interquartile Range
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **Watch for this mistake:** *A common mistake in Display Data: Box Plots is treating a quartile as just one data point instead of the median of a half — for the lower half 4, 8, 10, 14, some students pick 10 (the last value) as Q1 instead of finding the median of that half: Q1 = (8 + 10) ÷ 2 = 9. A second common slip is finding IQR by adding the quartiles instead of subtracting, writing IQR = Q3 + Q1 (like 30 + 18 = 48) instead of IQR = Q3 − Q1 (30 − 18 = 12) — always order the data first, then subtract Q1 from Q3 to find the spread of the middle 50%.*
8. **Try It 1:** A. Q1 = 12, Q3 = 28 — *With 8 values the median splits the data into a lower half (6, 10, 14, 18) and an upper half (22, 26, 30, 34). Q1 is the median of the lower half: (10 + 14) ÷ 2 = 12. Q3 is the median of the upper half: (26 + 30) ÷ 2 = 28.*
9. **Try It 2:** C. Between 15 and 30 — *The box of a box plot stretches from Q1 to Q3, and that box always holds the middle 50% of the data. Here the box goes from Q1 = 15 to Q3 = 30.*